

Juwon Kim

Postdoctoral Researcher, School of Electrical Engineering, KAIST
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Research Interests

Kernel, Filesystem, Flash storage, Device firmware (Flash Translation Layers), and Garbage collection.

Education

Ph.D. in Electrical Engineering Mar. 2023 – Aug. 2026
Korea Advanced Institute of Science and Technology (KAIST) Daejeon, Korea

- Advisor: Prof. Youjip Won

M.S. in Electrical Engineering Mar. 2021 – Feb. 2023
Korea Advanced Institute of Science and Technology (KAIST) Daejeon, Korea

- Thesis: *IPLFS: Log-Structured Filesystem without Garbage Collection*
- Advisor: Prof. Youjip Won

B.S. in Electrical Engineering Mar. 2016 – Feb. 2021
Korea Advanced Institute of Science and Technology (KAIST) Daejeon, Korea

- Individually Designed Major in Computer Science and Physics

Publications

- [1] **[ACM SOSP '26]** Juwon Kim, Hwanseok Yoo, Seung Won Yoo, and Youjip Won, “GAZE: Rearchitecting Log-Structured Filesystem for Garbage Collection on Zoned Namespace Flash Storage”, accepted for publication in *Proc. of ACM Symposium on Operating Systems Principles (SOSP)* (acceptance rate: 15%), Prague, Czechia, Sep.–Oct. 2026.
- [2] **[IEEE NVMSA '26]** Jihoon Seok, Juwon Kim, Youjip Won, “LeaST: Learned Sector Translation for Skipped Physical Page Address”, in *Proc. of IEEE Non-Volatile Memory Systems and Applications Symposium (NVMSA)*, Aug. 2026.
- [3] **[USENIX FAST '25]** Juwon Kim, Seungjae Lee, Joontaek Oh, Dongkun Shin, Youjip Won, “D2FS: Device-Driven Filesystem Garbage Collection”, in *Proc. of USENIX Conference on File and Storage Technologies (FAST)* (acceptance rate: 21%), Santa Clara, USA, Feb. 2025.
- [4] **[USENIX FAST '25]** Jieun Kim, Joontaek Oh, Juwon Kim, Seung Won Yoo, Youjip Won, “OPIMQ: Order Preserving IO Stack for Multi-Queue Block Device”, in *Proc. of USENIX Conference on File and Storage Technologies (FAST)* (acceptance rate: 21%), Santa Clara, USA, Feb. 2025.
- [5] **[ACM SAC '25]** Juwon Kim, Dongeon Kim, Seung Won Yoo, Myoengin Cheon, Joontaek Oh, Youjip Won, “lwFSCK: Light-weight Filesystem Check”, in *Proc. of ACM Symposium on Applied Computing (SAC)*, Catania, Italy, Mar. 2025.
- [6] **[USENIX ATC '22]** Juwon Kim, Minsu Jang, Danish Muhammad Teeshen, Joontaek Oh, Youjip Won, “IPLFS: Log-Structured File System without Garbage Collection”, in *Proc. of USENIX Annual Technical Conference (ATC)* (acceptance rate: 16%), San Diego, USA, Jul. 2022.

Posters and Workshops

- [1] **[ACM HotStorage '26]** Hwanseok Yoo, Juwon Kim, Youjip Won, “Manycore Scalable Logging for Log-Structured Filesystem,” in *Proc. of ACM Workshop on Hot Topics in Storage and*

File Systems (HotStorage), Prague, Czechia, Sep. 2026.

- [2] **[ACM SOSP '25] Juwon Kim**, Hwanseok Yoo, Youjip Won, “Accelerating Filesystem Garbage Collection for Zoned Flash Storage,” in *Proc. of ACM Symposium on Operating Systems Principles (SOSP)*, Seoul, Republic of Korea, Oct. 2025 (**Poster**).
- [3] **[ACM SOSP '25] Hwanseok Yoo, Juwon Kim**, Youjip Won, “Manycore Scalability of Modern Log-Structured Filesystem: Revisited,” in *Proc. of ACM Symposium on Operating Systems Principles (SOSP)*, Seoul, Republic of Korea, Oct. 2025 (**Poster**).
- [4] **[USENIX FAST '22] Juwon Kim**, Minsu Jang, Danish Muhammad Teeshen, Joontaek Oh, Youjip Won, “IPLFS: Log-Structured File System without Garbage Collection,” in *Proc. of USENIX Conference on File and Storage Technologies (FAST)*, Santa Clara, USA, Feb. 2022 (**Poster**).

Patents

- [1] **Juwon Kim** and Youjip Won, “Storage Device Based on Zoned Nanespaces, Host Controlling the Storage Device, and Storage System Including the Same,” Korean Patent Application No. 10-2026-0166149, filed Sep. 2026.
- [2] Youjip Won and **Juwon Kim**, “Dedicated Page Pool and In-Transit Data Consistency Management for Filesystem Garbage Collection,” Korean Patent Application No. 10-2026-0130283, filed Jul. 2026.
- [3] Youjip Won and **Juwon Kim**, “Selective Admission of Buffered Writes During Filesystem Garbage Collection,” Korean Patent Application No. 10-2026-0130284, filed Jul. 2026.
- [4] Youjip Won and **Juwon Kim**, “Method and Device for Managing Host Area,” Korean Patent Application No. 10-2025-0174419, filed Nov. 2025.
- [5] **Juwon Kim** and Youjip Won, “Storage Device Performing Garbage Collection,” Korean Patent Application No. 10-2025-0090393, filed Jul. 2025. (*Joint filing: SK Hynix & KAIST*)

Awards & Honors

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| • Student Scholarship, ACM SOSP, Czechia | Sep. 2026 |
| • Student Grant (Full), USENIX FAST, USA | Feb. 2025 |
| • NVRAMOS 2023 OpenSSD Best Practice Award, Gold Prize, KIISE, Korea | Nov. 2023 |
| • Student Grant (Full), USENIX FAST, USA | Feb. 2022 |

Research Experience

Postdoctoral Researcher Operating Systems Laboratory, KAIST	<i>Sep. 2026 – Present</i> Daejeon, Korea
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Visiting Researcher SK Hynix	<i>Jul. 2025 – Aug. 2025</i> Bundang, Korea
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Filesystem Garbage Collection for Zoned Mobile Storage

- Investigated filesystem garbage collection overhead on mobile platforms and explored host-side optimizations for zoned UFS devices.

Zoned UFS, F2FS, Garbage collection, Mobile storage, Linux kernel

Graduate Researcher (Ph.D. + M.S.) Operating Systems Laboratory, KAIST	<i>Mar. 2021 – Aug. 2026</i> Daejeon, Korea
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Log-Structured Filesystem for Large-Zone Reclamation

- Rearchitected a log-structured filesystem for efficient reclamation of large zones in Zoned Namespace flash storage, eliminating fundamental CPU and I/O bottlenecks in conventional

filesystem garbage collection.

F2FS, Zoned Namespace SSD, Garbage collection, Linux kernel, Memory management

Device-Driven Filesystem Garbage Collection

- Designed a host-device cooperative filesystem architecture that delegates filesystem garbage collection to the storage device, reducing host CPU utilization and host-device data movement.

Filesystem, Flash translation layer, Garbage collection, Host-device cooperation, NVMe

Log-Structured Filesystem without Garbage Collection

- Designed a log-structured filesystem that eliminates filesystem-level garbage collection by virtually expanding the filesystem partition over additional physical storage space.

F2FS, Log-structured filesystem, Garbage collection, Flash storage, Linux kernel

Undergraduate Research Intern

Jan. 2020 – Feb. 2021

Operating Systems Laboratory, KAIST

Daejeon, Korea

Log-Structured Filesystem Design

- Investigated log-structured filesystem design and filesystem garbage collection mechanisms under Prof. Youjip Won.

Filesystem, F2FS, Linux kernel, Flash storage

Work Experience

Machine Learning Engineer Intern

Jun. 2018 – Jan. 2019

Penta Security Systems Inc.

Seoul, Korea

CAN Data-Based Anomaly Detection for Electric Vehicles

- Designed and implemented a GRU-D autoencoder for detecting anomalous vehicle behavior from CAN bus data.

Machine learning, GRU-D, Autoencoder, CAN bus, Anomaly detection ““

Research Projects

Ultra-Low-Latency Storage-Driven I/O Subsystem for Large Language Models

Jul.

2026 – Present

InnoCORE, Ministry of Science and ICT, Korea

Order Preserving Distributed Storage System

Jun. 2025 – Present

NRF, Korea

Optimizing Filesystem Garbage Collection for Zoned Flash Storage

Sep. 2024 – Present

SNU-SKH Solution Research Center, SK Hynix, Korea

Device-Driven Filesystem Garbage Collection

Sep. 2023 – Aug. 2024

SNU-SKH Solution Research Center, SK Hynix, Korea

Collection and Integrated Analysis of Automotive Inter/Intra System Artifacts

Jan.

2024 – Feb. 2025

IITP, Korea

Future Scalable OS: Log-Structured Filesystem without Garbage Collection

Jan.

2021 – May 2023

IITP, Korea

Teaching Experience

- Head TA, Operating Systems and System Programming for EE (EE415), KAIST Sep.–Dec. 2025

- TA, Special Topics in Computer Engineering (EE817), KAIST Feb.–Jun. 2025

- TA, Operating Systems and System Programming for EE (EE415), KAIST Sep.–Dec. 2024
- Head TA, Programming Structure for EE (EE209), KAIST Feb.–Jun. 2024
- TA, Programming Structure for EE (EE209), KAIST Sep.–Dec. 2023
- TA, Programming Structure for EE (EE209), KAIST Feb.–Jun. 2023

Service

- Artifact Evaluation Committee, USENIX ATC '26 Sep. 2026 – Oct. 2026
- Artifact Evaluation Committee, USENIX FAST '27 Spring Cycle Jun. 2026 – Jul. 2026
- Lab Manager, Operating Systems Laboratory, KAIST Jan. 2026 – Jul. 2026
- Program Chair Assistant, USENIX FAST '24 Sep. 2023 – Sep. 2024

Technical Skills

Systems: F2FS, flash translation layer, SSD firmware, NVMe driver, block I/O layer, ZNS, Linux page management subsystem

Languages: C, Python, C++, SQL, Verilog

Tools: git, vim, gnuplot, ctags, cscope, ftrace, blktrace